

Good news: Team Rocket finally got their Pikachu cloning machine working. Bad news: They don't know how to turn it off.

Help the local area Pokémon authorities figure out how many trainers they need to summon to catch the run-away pikachus before they overrun the city, by figuring out the cloning sequence being used!

## Input

You will receive a sequence of integers on a single line, separated by commas.

20, 26, 32, 38, 44, 50, 56, 62

## Output

Determine the sequence and the next number in the sequence, and print them out one per line. If the sequence cannot be determined write UNKNOWN for both outputs. If the sequence is addition output a "+" in front of the sequence, for subtraction output "-", for multiplication "\*" and for division "/"

Sequence: +6  
Next: 68



## Discussion

The sequence can be positive or negative, and can involve Multiplication, Division, Addition or Subtraction. Only whole number sequences will be given in the datasets. The same single operation will be done on every sequence step (e.g. the sequence will not be something like \*8, then +3 in one step). Each step will be the same INTEGER change (\*Integer, -Integer, etc.) Division will result in integers, and any division floating point points generated when calculating the next number in the sequence need to be converted to integers.

Reminder: have you run your solution against **all** of the student data sets?

## Additional Examples

| Input 2                 | Output 2                 | Input 3                         | Output 3                           |
|-------------------------|--------------------------|---------------------------------|------------------------------------|
| 95, 90, 85, 80, 75, 70  | Sequence: -5<br>Next: 65 | 3, 30, 300, 3000, 30000, 300000 | Sequence: *10<br>Next: 3000000     |
| Input 4                 | Output 4                 | Input 5                         | Output 5                           |
| 1024, 512, 256, 128, 64 | Sequence: /2<br>Next: 32 | 18, 32, 2, 97, 100, 1, 15       | Sequence: UNKNOWN<br>Next: UNKNOWN |